

Chapter 1

Computer-assisted Language Mediation in Teaching Human-centred Augmented Translation

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In the evolving landscape of translation education, the integration of computer-assisted language mediation has emerged as a key strategy for teaching human-centred augmented translation. This study explores the implementation of generative AI (GenAI) models, in particular Large Language Models (LLMs), in translation workflows to enhance translator competence while maintaining ethical standards. Using custom AI systems that adapt and evolve with new translations, students can be trained to critically evaluate AI-generated content and manage hybrid translation workflows that combine AI output with human expertise. The research emphasises the importance of understanding the limitations and potential biases of AI, and advocates a balanced approach in which AI augments rather than replaces human judgement. Practical examples and exercises demonstrate the strengths and weaknesses of using AI in information processing and multilingual text generation, with the ultimate aim of training a new generation of translators to use AI technologies responsibly.

1 Introduction

1.1 GenAI in translation education

In translation education, integrating generative AI (GenAI) models into translation processes presents challenges in maintaining translator autonomy while embracing technology (O'Brien2024). Ethical concerns, including privacy, algorithmic bias, and responsible use of machine translation and linguistic data,

must be carefully considered as translation education undergoes a paradigm shift (Ramírez-PoloEtAl2023, SlimiEtAl2023, TavaresEtAl2023).

Developing computer-assisted language mediation in translation education requires successfully integrating data analysis and AI assistants into pedagogical workflows. In this respect, AI is one of the building blocks of translation technologies, along with computer-assisted translation environments, project management software, ontologies, expert involvement and data management facilities (Garcia2023, Mitchell-Schuitevoerder2020, HeEtAl2023).

1.2 Computer-assisted language mediation

Used as part of computer-assisted dynamic mediation framework (HuEtAl2023), GenAI tools allow translation students not only to produce multilingual content, but also to compare and fine-tune the generated output, fostering a deeper contextual understanding of the strengths and weaknesses of different translation models and enabling future translation professionals to become more proficient reviewers of automatically generated content (ZhuangEtAl2023). This process requires new translation skills and up-to-date knowledge of AI assistants built on underlying AI models, such as Large Language Models (LLMs) (Douglas2023).

As LLM technology is introduced to the field of translation, enabling trainees to navigate language variations within adaptive pathways, the concept of a linguistic norm emerges as an increasingly crucial analytical tool, facilitating efficient translation across diverse contexts (Fabricius2022, SchulzOllig2023, Sinner2020). From this perspective, computer-assisted language mediation can be taught as a set of new translation competences, including the ability to create and use a language assistant, and potentially to understand how to employ LLM-enhanced linguistic data responsibly (RazaEtAl2024).

From a technical standpoint, building personalised AI assistants from scratch requires substantial resources, emphasising the efficiency of leveraging pre-trained models (ShichkinaKrinkin2022). This process can be part of training programs specifically tailored for cultivating data scientists specialised in translation. Few-shot learning (FSL)¹, particularly in-context learning (ICL)², empow-

¹ Few-Shot Learning (FSL) is a machine learning approach that enables models to adapt to new tasks with only a small amount of labelled data (MosbachEtAl2023). For more information: <https://medium.com/ubiai-nlp/step-by-step-guide-to-mastering-few-shot-learning-a673054167a0>

² In-context learning, as popularised in large language models such as GPT-3 and GPT-4, involves a model's capacity to comprehend and execute tasks based on contextual information embedded within the input sequence, without modifying the model's parameters (DongEtAl2023).

ers translators with limited natural language processing (NLP) expertise. This approach allows LLMs to rapidly acquire new capabilities, thereby facilitating seamless integration of AI into a range of translation workflows. In this regard, prompt engineering³ facilitates the development of a versatile approach that can be applied to a range of tasks, including multilingual translation, fine-tuning of translation models (DogruMoorkens2024), information extraction, and text summarisation. Training future translators to effectively prompt LLMs for these tasks in digital communication is essential. Our analysis in Section ?? will provide specific examples that illustrate these approaches in teaching.

1.3 Investigating biases and inconsistencies of GenAI

An important area of research in teaching GenAI technologies in translation courses is to make students aware of the potential biases and inconsistencies they may encounter when using this new generation of tools. The shortcomings of AI-generated content become apparent when the translation task is fully mastered. Unfortunately, if students are not fully proficient in the target language and lack specialised knowledge and translation practice, they may find it difficult to effectively assess the quality of the output (TavaresEtAl2023). Understanding both the strengths and limitations of GenAI through a series of practical examples is vital for efficient use of translation tools (IULMFarrell2023).

By exposing students to real-life scenarios and practical exercises, it is possible to develop a better understanding of the qualities and shortcomings of automatically generated content in the context of multilingual communication. This hands-on approach allows trainee translators to identify potential biases, inconsistencies and errors that may arise from relying solely on AI assistance. In addition, by analysing and discussing these practical examples, students can learn to critically evaluate the output of AI assistants and identify when human intervention or additional expertise is required to produce accurate and contextually relevant content. This knowledge is also essential for developing a new generation of translation-revision workflows that successfully combine GenAI and human expertise.

³The field of prompt engineering is concerned with the design and optimisation of prompts to help a model perform better on new tasks with limited data. For prompts to be effective, they must clearly define the task at hand, include context to help the model use its existing knowledge, and gradually introduce new information or tasks.

2 Contextual understanding and automation

Artificial intelligence is now capable of processing a wide variety of information, including text, images, speech, facial expressions, and, to a certain extent, body movements. These capabilities can be used to develop a large panel of AI assistants (YangEtAl2024). One of the most intriguing aspects of these technologies is their handling of context, which differs significantly from human understanding (O'Brien2024, ZhuEtAl2024).

Human understanding involves a combination of experience, emotion, cultural background and situational awareness. This allows people to naturally interpret nuances, implicit meanings and subtleties in communication. Assessing human understanding of context through written expression has long been a focus of research, with methodologies evolving to encompass cognitive, linguistic, and technological perspectives (O'Brien2024). Researchers in cognitive psychology and linguistics have devoted considerable time and effort to studying the mechanisms of language processing, with a particular focus on the role of lexical access, syntactic parsing and discourse coherence in comprehension. In addition, discourse analysis has provided valuable insights into how contextual cues shape interpretation, revealing complex interplay between language and cognition in human comprehension (Van Dijk2006, PleyerWinters2015, GledhillPecman2018).

In contrast, GenAI models process context through the identification of patterns within the data they have learned during the training phase (ZhuangEtAl2023). In contrast to a true understanding of context, these models rely on vast quantities of data to identify and generate responses based on statistical correlations. To illustrate, when presented with a sentence, an AI assistant analyses the sequence of words and predicts the most probable continuation based on the patterns it has learned. Currently, this approach can effectively imitate human-like responses, but it lacks genuine comprehension and the capacity to grasp more profound meanings or unexpected cultural nuances. Additionally, it tends to accentuate recognised patterns in the generated output.

This difference in contextual understanding has significant implications for translation. While AI may be good at generating coherent and contextually appropriate text, it may struggle with tasks that require cultural sensitivity or complex problem-solving that requires nuanced human judgement. Furthermore, AI's understanding of context is limited to the data on which it has been trained. If training data sets lack diversity or are biased, AI-generated content will reflect these limitations (MartínezEtAl2023). Understanding these differences is crucial when integrating GenAI assistants into translation workflows. While AI-based technologies can significantly increase productivity and efficiency by performing

routine tasks and providing quick information, they should complement rather than replace human judgement and expertise in translating certain types of content (DenningArquilla2022).

Balancing the strengths of AI in processing and generating text with advanced human skills used to understand and interpret complex contexts can lead to more efficient and nuanced applications of translation technology for translation project management, known as HUMAN-CENTRED AUGMENTED TRANSLATION (O'Brien2024).

In practical terms, what specific tasks can help students understand GenAI weaknesses as reflected in the warning “ChatGPT can make mistakes. Check important info.” (<https://chatgpt.com>)? In what types of situations are these tools helpful, and what are the potential pitfalls? The following section explores the point of studying biases and inconsistencies in GenAI operation through practical examples of classroom activities.

3 Exploring AI in practice: delving into knowledge-rich texts

3.1 Text summarisation and interactive learning with AI assistants

In this section, we consider how practical tasks can help students gain a deeper understanding of GenAI. One of the easiest ways to address these complex processes in translation education is to use examples of knowledge-rich texts (GledhillKübler2016). In this context, information synthesis is one of the key capabilities that AI brings to streamline information processing, with tools such as SciSpace and PDFgear being able to summarise key information almost instantly for efficient retrieval.⁴ In this respect, linguistic standards, specifications and style guides provide an excellent basis for practice (GledhillZimina-Poirot2023). These well-structured documents precisely outline the standards that should be applied in written communication. At the same time, the length and complexity of these documents often require detailed human analysis and extensive practice to fully master the rules.

To illustrate how AI can be used in teaching, let us consider a specific example. In a Master's level 2 course on technical translation and controlled languages, students are expected to familiarise themselves with ASD-STE100 (Simplified

⁴ SciSpace is an advanced platform designed for researchers that uses natural language processing and machine learning to facilitate efficient literature review and analysis in scientific fields.

Technical English).⁵ ASD-STE100 is an international specification for writing technical documentation in a controlled natural language.⁶ This comprehensive document of over four hundred pages requires considerable effort to master the principles of STE for writing procedures, descriptive technical texts, warnings and cautions. In addition, there is an extensive “core dictionary” containing a wide range of lexical items that students must learn to use in context. The dictionary is accompanied by illustrative examples and contrasting cases (APPROVED: STE/not approved: non-STE).

Integrating AI into ASD-STE teaching can potentially improve learning experience. The following examples highlight several ways in which this can be achieved.

- Use case: Use tools such as PDFGear or SciSpace to quickly summarise lengthy sections of ASD-STE100.
- Benefit: By using a concise format in classroom settings with limited teaching time, students can focus on understanding and practical applications of the STE rules rather than being overwhelmed by the volume of text.

In this context, AI assistants can act as interactive tutors, answering students’ questions about specific language policies and providing immediate feedback. Prompt clarification and reinforcement can accelerate learning. In addition, AI assistants can quickly locate specific sections or rules within lengthy documents through enhanced research and reference, saving time and improving efficiency. The main benefit is that students can spend more time applying the rules and less time searching for them.

At the same time, through this type of exercise, students will realise that both PDFGear and SciSpace copilots are quite effective at extracting relevant knowledge and providing a comprehensive list of essential rules to be applied in writing. Nevertheless, this knowledge extraction is based on patterns in the text and may be superficial when it comes to applying these rules in writing and rewriting texts. Students can rapidly become aware of these limitations and gain insight

⁵ See curriculum for Master 2 ILTS, Université Paris: <https://odf.u-paris.fr/fr/offre-de-formation/master-XB/arts-lettres-langues-ALL/traduction-interpretation-K6JMSAFS/master-traduction-interpretation-parcours-industrie-de-la-langue-et-traduction-specialisee-JRQNFV5Z.html>

⁶ ASD-STE100 (<https://www.asd-ste100.org>) is a set of writing rules and a controlled vocabulary for producing clear and simplified technical documentation in English. It was developed in the 1980s by the aerospace industry associations (AECMA/ASD and AIA) to make aircraft maintenance documentation more readable for people with limited English proficiency.

into the technical aspects of producing content with GenAI. This dual awareness can help them to understand both the potential and the limitations of using AI assistants in the context of technical communication and translation:

- Use case: Use PDFgear and SciSpace copilots to generate examples and counter-examples based on the core dictionary and ASD-STE100 rules, then critically evaluate the AI's output.
- Benefits: Firstly, this type of exercise encourages active learning and critical thinking as students compare AI-generated content with examples of human rewriting. Secondly, it provides insight into the limitations of automatic text generation in the practical application of writing rules.

The following section will demonstrate how previous exercises can be further developed into a more conclusive experiment, using tools such as ChatGPT, Perplexity AI, or Claude.ai.⁷ In this case, a prompt language model (input in Technical English and output in Simplified Technical English) would be employed to demonstrate the outcomes of few-shot learning on a specific data set. In addition, prompts can be used to highlight inconsistencies in the generated output, thereby directing attention towards potential enhancements (such as the avoidance of noun clusters and the use of articles encouraged by STE).

3.2 AI output refinement using prompt engineering and few-shot learning

From a technical perspective, building LLMs from scratch requires significant resources, which highlights the efficiency of using pre-trained models. This process can be integrated into teaching programs specifically designed to train data scientists specialising in translation. Few-shot learning, especially in-context learning, enables translators with limited NLP expertise to quickly acquire new skills and seamlessly integrate AI into translation workflows across multiple applications.

In the context of few-shot learning, a model is presented with a limited number of labelled examples corresponding to a desired input-output scenario. Transfer learning principles are embedded in the model to enable it to use these examples

⁷ ChatGPT is a conversational AI model developed by OpenAI based on the GPT (Generative Pre-trained Transformer) architecture. Based on OpenAI's model, and a standalone LLM with NLP capabilities, Perplexity AI is a conversational search engine that answers queries using natural language predictive text, using sources from the web. Claude.ai is trained by Anthropic using Constitutional AI (BaiEtAl2022).

to adjust its parameters and fine-tune its representations to enhance the presence of specific patterns. The main advantage of few-shot learning is its ability to generalise from limited examples to unseen data, due to the rich abstract representations acquired during pre-training. However, even in this scenario, the results may not be entirely satisfactory and may require human refinement.

- Use case: Create a prompt language model with labelled examples of “Input text in Technical English” and “Output text in STE”, demonstrating the results of few-shot learning on a given data sample.
- Benefit: This hands-on exercise allows students to interact directly with the technology and gain practical experience using prompt language models for text generation tasks.

In addition, this approach provides an opportunity for students to observe the limitations of AI in fully capturing complex linguistic nuances, highlighting the importance of human analysis and editing to refine the output for satisfactory results.

3.3 Terminology extraction with LLMs: understanding model limitations

In parallel, other initiatives are exploiting the capabilities of combining AI and NLP to facilitate access to specialised knowledge. Teaching terminology extraction from multimodal sources using LLMs is one of the areas which can reinforce learners’ augmented reasoning in translation.

Before using LLMs with multimodal capabilities (such as Perplexity AI playground: <https://labs.perplexity.ai> or <https://gemini.google.com>) for the analysis and extraction of terms from multiple sources, it is beneficial for students to contrast the outcomes generated by LLMs with those produced by conventional term extraction, text-mining, and text annotation tools. This comparison enables learners to become aware of potential biases and hallucinations. For example, the frequencies of candidate terms identified by GenAI are not always 100% accurate compared to traditional text segmentation and term extraction methods. AI models can propagate errors from misinterpretations in the text, resulting in inaccurate term frequencies. Misreading sentence structure can lead to incorrect counts of certain terms (Uchida2024).

In addition, AI models trained on diverse and large datasets may inappropriately apply patterns learned in one context to another. For example, if a model

has frequently seen “climate change” discussed alongside terms such as “policy” and “regulation”, it might overestimate the frequency of these terms in any document mentioning “climate change”, even if the specific text focuses on scientific data rather than policy.

These errors highlight a key limitation of AI models: their reliance on patterns and associations learned from vast amounts of data, which can sometimes lead to misinterpretations if the specific context of a new text differs from those patterns. The generated output exaggerates the learned patterns. As a result, the frequency of terms identified by GenAI may not always accurately reflect the actual text and its context, especially in cases where a nuanced understanding of language is required. Therefore, comparing AI-generated results with those of traditional text segmentation and term extraction tools, which often use straightforward statistical methods, can help identify and correct such inaccuracies.

- Use case: Use LLMs to extract terms from different types of documents. Check the extracted terms for accuracy and relevance. Use text-mining and navigation tools (such as Voyant Tools: <https://voyant-tools.org>) to critically evaluate the results produced by AI assistants.
- Benefit: Learn to use quantitative methods to evaluate the quality of AI-generated text analysis.

Note that ready-made prompts for text analysis can be found on many websites.⁸ Such models can be adapted, for example, for term extraction (works for Perplexity AI):

- Use case (sample prompt): Extract the most relevant terms with contexts from the attached document. Present the results in a markdown table with the following columns: term, frequency, contexts, definition.⁹

3.4 Towards “augmented reasoning” in translation: linguistic data and LLMs

Many other experiments with AI tools are possible in the classroom, both to demonstrate the role that AI models can play in different application settings,

⁸ For example, Keywords Everywhere ChatGPT Prompt Templates: <https://keywordseverywhere.com/chatgpt-prompt-templates.html> (also available for Claude.ai).

⁹More examples are available in #2024TEF -AI-powered terminology extraction: A hands-on guide for translators by Josh Goldsmith: <https://youtu.be/5Y5PhzyeMGI?list=PLLqIRaiVCGCR80tysPOQ2AHJ30HEyItXX>

such as text simplification, paraphrasing, answering questions, and generating synthetic data to automate translation tasks. Furthermore, students can conduct experiments to explore and address biases in word embeddings¹⁰, which play a pivotal role in the functioning of LLMs.

In addition, corpus linguistic tools and methods can be used to identify recurrent characteristic patterns in source language texts and comparable target language texts. This approach enhances the performance of LLM-based translation assistants by leveraging targeted datasets containing these characteristic elements. For example, translation students can be trained to identify comparable sentences in multilingual annotated corpora (treebanks) and learn how to prompt LLMs to translate these sentences correctly within a given linguistic context.

Moreover, in this scenario, human-centred information retrieval becomes essential for identifying comparable texts in the target language. To support this, corpus linguistic tools and methods, such as co-occurrence network analysis and syntactic dependency annotation, can be employed to extract specific patterns that capture terminological, phraseological, and syntactic features. These patterns are then integrated into LLM training for translation purposes, potentially through the implementation of a translation bot, such as one designed to write bilingual definitions.

Developing such skills requires creativity and non-standard thinking, offering many potential applications in the language industry, particularly in managing data intelligence platforms¹¹. Consequently, GenAI can become a crucial component of translation technology. However, it must be carefully controlled by experts and combined with reliable data management facilities.

Finally, a promising area of educational research lies in the development of augmented human translation in translation pedagogy. This approach aims to foster a new competence called ‘augmented reasoning’ in translation, which combines human intuition with robust data support to optimise translation processes.¹²

¹⁰ An embedding is a vector representation of data designed to capture and preserve aspects of its content or meaning. For more information on bias detection in word embeddings and LLMs for European languages: https://www.biasproject.eu/scientific_publications/the-bias-detection-framework-bias-detection-in-word-embeddings-and-language-models-for-european-languages

¹¹ Databricks: <https://www.databricks.com>

¹² EMT competence framework 2022: https://commission.europa.eu/education/skills-and-qualifications/develop-your-language-skills/european-masters-translation-emt/european-masters-translation-emt-explained_en

4 Hybrid translation workflows: enhancing human expertise with GenAI

4.1 Augmented cognition: challenges in translation education

In translation education, the integration of LLMs into translation poses challenges in terms of maintaining learner autonomy and developing translation competence while taking advantage of technological advances. Maintaining learner autonomy means emphasising that GenAI assistants should be seen as tools, rather than as a substitute for human translators. Learners should retain control over the translation process and engage in critical evaluation of the output generated by automated translation systems, thereby developing skills in assessing quality and appropriateness (GledhillZimina-Poirot2023). Furthermore, learners must learn to provide clear instructions to LLMs, which is crucial for effective human-machine collaboration (Dr Kantharaja K P et al. 2024).

As translation education undergoes a paradigm shift, it is becoming increasingly important to encourage responsible use of machine translation and linguistic data, with careful consideration of ethical concerns such as privacy, data education and algorithmic bias. To ensure effective use of LLMs in translation processes, it is essential to adopt a comprehensive approach, including seamless integration of technology into pedagogical workflows through computer-assisted language mediation.

In the past, efficient interaction with machines and data navigation techniques also relied on human intuition backed up by data, but implied a clear distinction between quantitative information, which can be measured and counted, and human judgement (qualitative analysis) (ZiminaFleury2016). The realisation of augmented cognition requires a stronger connection between the user and the system, which should progressively learn to understand human functioning and respond effectively to situational demands. From this perspective, personalised AI plays an important role in facilitating this integration.

The convergence of human-machine relationships is a major challenge for translation professionals, spanning academic disciplines such as translation studies and the translation industry (Raus2023). The emphasis on augmenting intelligence rather than replacing humans marks a shift from emulation to empowerment, which is a progressive way forward.

In the industry, the foundations of this approach lie in hybrid translation workflows that seamlessly combine human expertise with AI capabilities, rather than relying on a distinct separation between qualitative analysis and automatically generated content, which is then edited and revised by a human expert. In the

coming years, acquiring expertise in the design of advanced translation workflows that integrate innovative methods and technologies will become a key competence in Master's level translation education. In the context of industrial applications, tools that facilitate such workflows, such as those available in the RWS Language Cloud¹³, will be enhanced with upcoming GenAI features. For example, RWS's linguistic AI solution EVOLVE aims at "integrating human and artificial intelligence, revolutionising translation processes and the time it takes to achieve high quality results".¹⁴

The following section offers a brief overview of the principles behind a hybrid translation workflow that integrates elements of Dynamic Translation Memory (DTM) and leverages dynamic human-AI interaction in project-based translation.

4.2 Dynamic Translation Memory (DTM) in project-based translation

Unlike traditional Translation Memory (TM)¹⁵, a Dynamic Translation Memory (DTM) is part of custom AI systems that not only store translations, but also adapt and evolve, learning from new translations and adapting to different contexts within a project-oriented translation workflow (HaoEtAl2023, KimEtAl2022, Sin-wai2023, Zimina2015, ZiminaFleury2016, Zimina2019).

Recent developments in LLMs have demonstrated their capacity to outperform traditional neural machine translation models in certain contexts and language combinations. For example, a Lionbridge assessment (2023)¹⁶ showed that certain GenAI models can perform better than neural machine translation engines, challenging the established paradigm in this sector. At the same time, combining neural machine translation with neural fuzzy adaptation (NFA) is a promising way to achieve flexible control over translation resources while avoiding potential biases inherent in general purpose LLMs.¹⁷

Given these technical advances, a DTM system based on AI technologies differs from traditional TM systems in that it not only stores translated text, but also adapts and evolves in response to new translations and changing contexts within an AI-enabled translation workflow. The development of new translation management skills is therefore essential in this evolving landscape.

¹³ <https://www.rws.com/localization/language-cloud>

¹⁴ <https://www.rws.com/fr/about/news/2024/evolve-by-RWS-transforms-global-translation-practices/>

¹⁵ <https://www.trados.com/learning/topic/translation-memory>

¹⁶ <https://www.lionbridge.com/fr/blog/translation-localization/machine-translation-a-generative-ai-model-outperformed-a-neural-machine-translation-engine>

¹⁷ <https://www.systransoft.com/translation-products/translation-memories>

To illustrate some practical applications of this approach in teaching project-based translation at the Master's level, a number of specific examples are required. Let's consider the task of implementing a dynamic workflow where both TM and LLM interact. The following examples illustrate the range of related processes and tools/services that are currently available on the market for designing a dynamic workflow:

- Leveraging TM in a dynamic workflow (segment analysis). For each segment of the source text, the TM software checks for matches. If an exact or high confidence fuzzy match is found, it is such as user dictionaries and TM (available, for example, on the SYSTRAN Translation Server) (for practical examples: [Zimina-PoirotGledhill2023](#)).
- If a fine-tuned translation model is not available, it is possible to use LLM integration with Adaptive Generative Translation (AGT).¹⁸ For segments without customised NMT output, LLM integration allows a translation to be generated. This can be done in real-time or batch mode. For example, memoQ AGT uses the in-context learning capability of LLMs to generate more consistent translations using in-context resources (TM and glossary).¹⁹
- Translation consistency checks. LLMs can be used to ensure consistency across multiple documents or versions. In this case, translated texts (bi-texts) can be fed into AI assistants, such as Poe (<https://poe.com>), to analyse terminology, phrasing, and style consistency. Prompt example: Identify and resolve inconsistencies to ensure consistency across translations. Produce a report highlighting areas of inconsistency and providing recommendations for improvement.
- Human post-editing and revision. Translators review, post-edit and revise the combined output to ensure consistency and quality. At this stage, an assessment report quantifies the revisions and highlights the strengths and weaknesses of the translated content.²⁰
- Overall translation quality assessment and evaluation. Various assessment techniques are employed in the industry for translation evaluation, and these methods are increasingly enhanced with AI-based features.²¹

¹⁸ <https://agt.memoq.com>

¹⁹ <https://www.memoq.com/resources/webinars/memoq-agt>

²⁰ <https://www.rws.com/localization/services/language-quality-assurance>

²¹ A good starting point for an analytic summary can be found here: <https://videotap.com/blog/7-translation-quality-assessment-techniques>

These industrial practices are key building blocks of a hybrid workflow that leverages AI-suggested output to enhance translation project consistency and efficiency. However, the true success of such project workflows lies in seamless integration of machine-driven efficiency and human oversight. In this respect, revision emerges as a crucial phase in ensuring semantic integrity and contextual relevance. While AI systems can flag inconsistencies, suggest improvements, and support adherence to glossaries and style guides, human intervention is essential to complement these tools. This evolving landscape highlights the importance of equipping trainee translators with both revision skills and the ability to design and implement workflows that ensure high-quality output at every stage.

4.3 Leveraging revision history in AI-assisted translation training

Trainees must actively participate in project management, navigating the intricate cycles of post-editing and revision while leveraging AI tools effectively. In a structured translation/revision workflow, the revision process is ‘evolutive’, meaning that modifications should not be random. With the exception of occasional edits, changes should result from a complex interplay of various editing constraints (for example, writing in Plain Language). This underscores an ‘evolutive’ perspective on the translation project: each stage of translation and revision must be carefully planned in advance, in collaboration with both clients and students (GledhillZimina-Poirot2023).

From this perspective, a revision history archive becomes a valuable resource in AI-assisted translation training, offering a comprehensive record of the edits and decisions made throughout the translation process. The analysis of revisions allows trainees to gain insights into the interaction between AI-generated suggestions and human intervention. This process helps students develop critical editing skills, such as identifying context-sensitive errors, improving terminology management, and refining stylistic consistency. Moreover, revision history enables instructors to provide targeted feedback by pinpointing recurring issues and showcasing best practices. A combined qualitative and quantitative analysis of these outputs reveals characteristic markers of the processes that contribute to the acquisition of the conventions specific to professional translation by the end of the project (Zimina-PoirotGledhill2023).

This structured approach not only enhances the learning experience but also fosters a deeper understanding of how to integrate AI tools effectively into real-world workflows, thus preparing students for the demands of professional translation projects. For example, Qualitative Translation/Revision Workflow (QTRW) teaching project design (Gledhill & Zimina-Poirot, 2023) is defined by its dual

responsibility of balancing tools workflow management and hands-on revision expertise. This approach has a potential to combine the advanced generative capabilities of different language models with human expertise, resulting in translations that are both consistent and contextually appropriate.

5 Preliminary conclusions and future work

It is clear that when GenAI assistants are used as part of project-oriented qualitative translation/revision processes, students become aware that artificial intelligence is only one of the tools at their disposal and that they also need to monitor and correct the results generated by AI models to ensure the quality of their work in accordance with each project step requirements. This approach can be time-consuming and requires significant human involvement, which may explain the presence of errors in multilingual documents generated by fully automated processes. In this context, there is a clear need for precise standards and a new paradigm for human-machine interaction.

When used as part of computer-assisted mediation, the revision history archives offer significant pedagogical potential. They allow students to compare translations, fostering a deeper understanding of contextual nuances while providing targeted guidance to help them become better translators. Moreover, integrating computer-assisted mediation into classroom activities is becoming easier as specific learning platforms become more widely available through educational portals.²² These advances will gradually enable translation educators to incorporate a variety of language resources and interactive content into their teaching practice.

As the technology continues to evolve, it is reasonable to expect the emergence of even more innovative multimodal AI services and predictive solutions that will continue to transform the translation technology landscape, facilitating interactive translation and project workflows with ‘augmented reasoning’ in mind: ENHANCING HUMAN EXPERTISE WITH AI.

²² CLARIN Learning Hub: <https://www.clarin.eu/content/learning-hubMultiTraiNMT>:
http://www.multitrainmt.euDataLitMT: <https://itmk.github.io/The-DataLitMT-Project>